

GRAPH MODELS FOR SPARSE MATRICES

- ▶ Sparse matrices and graphs.
- ▶ Paths and powers of matrices
- ▶ Perron Frobenius theorem
- ▶ Application: Markov chains
- ▶ Bipartite model, hypergraphs
- ▶ Application: back propagation

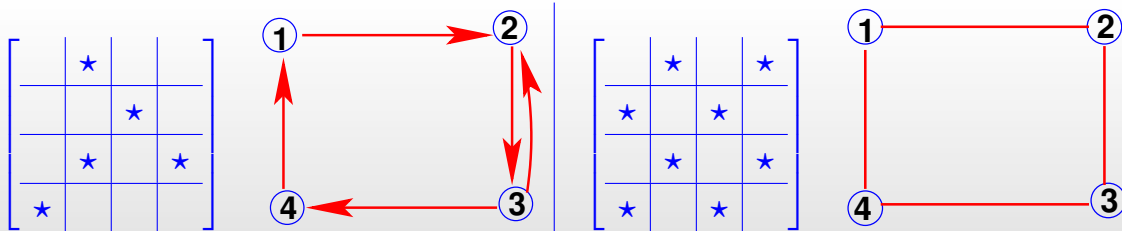
Graph Representations of Sparse Matrices. Recall:

Adjacency Graph $G = (V, E)$ of an $n \times n$ matrix A :

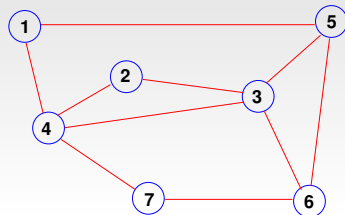
$$V = \{1, 2, \dots, N\} \quad E = \{(i, j) | a_{ij} \neq 0\}$$

◆ G == undirected if A has a symmetric pattern

Example:



 **1** Show the matrix pattern for the graph on the right and give an interpretation of the path v_4, v_2, v_3, v_5, v_1 on the matrix



✦ A separator is a set Y of vertices such that the graph G_{X-Y} is disconnected.

Example: $Y = \{v_3, v_4, v_5\}$ is a separator in the above figure



2 Adjacency graph of:

What is a 'good' separator for this graph?

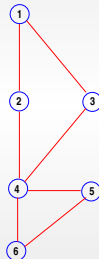
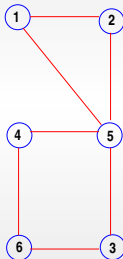
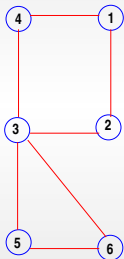
$$A = \begin{bmatrix} & \star & & \star & & \\ \star & & \star & & & \\ & \star & & \star & \star & \star \\ \star & & \star & & & \\ & & \star & & & \star \\ & & \star & & & \\ & & \star & & \star & \end{bmatrix}.$$



3 For any **adjacency** matrix A , what is the graph of A^2 ? [interpret in terms of paths in the graph of A]

- ▶ Two graphs are **isomorphic** if there is a mapping between the vertices of the two graphs that preserves adjacency.

 **4** Are the following 3 graphs isomorphic? If yes find the mappings between them.

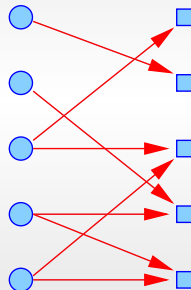


- ▶ Graphs are identical – labels are different
- ▶ Determining graph isomorphism is a **hard** problem

Bipartite graph representation

- ▶ Rows and columns are (both) represented by vertices;
- ▶ Relations only between rows and columns: Row i is connected to column j if $a_{ij} \neq 0$

Example:



✦ Bipartite models used only for specific cases [e.g. rectangular matrices, ...] - By default we use the standard definition of graphs.

Interpretation of graphs of matrices

 5 What is the graph of $A + B$ (for two $n \times n$ matrices)?

 6 What is the graph of A^T ?

 7 What is the graph of $A.B$?

 8 What is the graph of $A + B$ (for two $n \times n$ matrices)?

 9 What is the graph of A^T ?

 10 What is the graph of $A.B$?



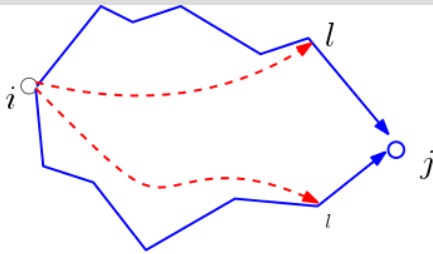
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What is the graph of A^k ? (Give answer in plain English)

Theorem

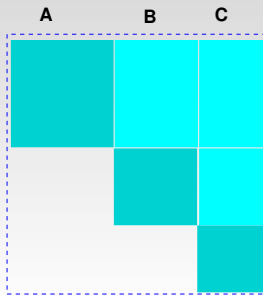
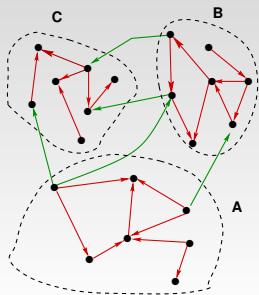
Let A be the adjacency matrix of a graph $G = (V, E)$. Then for $k \geq 0$ and vertices u and v of G , the number of paths of length k starting at u and ending at v is equal to $(A^k)_{u,v}$.

Proof: Proof is by induction. ■



If $C = BA$ then $c_{ij} = \sum_l b_{il}a_{lj}$. Take $B = A^{k-1}$ and use induction. Any path of length k is formed as a path of length $k-1$ to some node l completed by an edge from l to j . Because a_{lj} is one for that last edge, c_{ij} is just the sum of all possible paths of length k from i to j

- ▶ Recall (definition): A matrix is **reducible** if it can be permuted into a block upper triangular matrix.
- ▶ Note: A matrix is reducible iff its adjacency graph is not (strongly) connected, i.e., iff it has more than one connected component.
- ▶ Other definition: A is nonnegative if $a_{ij} \geq 0 \forall i, j$ (notation: $A \geq 0$)



✦ No edges from C to A or B . No edges from B to A .

Theorem: Perron-Frobenius An irreducible, nonnegative $n \times n$ matrix A has a real, positive eigenvalue λ_1 such that:

- (i) λ_1 is a simple eigenvalue of A ;
- (ii) λ_1 admits a positive eigenvector u_1 ; and
- (iii) $|\lambda_i| \leq \lambda_1$ for all other eigenvalues λ_i where $i > 1$.

✦ The spectral radius is equal to the eigenvalue λ_1

✦ Definition : a graph is d regular if each vertex has the same degree d .

Proposition

The spectral radius of a d regular graph is equal to d .

Proof The vector e of all ones is an eigenvector of A associated with the eigenvalue $\lambda = d$. In addition this eigenvalue is the largest possible (consider the infinity norm of A). Therefore e is the Perron-Frobenius vector u_1 . ■

Application: Markov Chains

- ◆ Read about Markov Chains in Sect. 10.9 of:

https://www-users.cs.umn.edu/~saad/eig_book_2ndEd.pdf

- ▶ Let $\pi \equiv$ row vector of stationary probabilities
- ▶ Then π satisfies the equation
- ▶ P is the probability transition matrix and it is 'stochastic':

$$\pi P = \pi$$

A matrix P is said to be stochastic if :

- (i) $p_{ij} \geq 0$ for all i, j
- (ii) $\sum_{j=1}^n p_{ij} = 1$ for $i = 1, \dots, n$
- (iii) No column of P is a zero column.

- ▶ Spectral radius is ≤ 1 . ✍ 12 Why?
- ▶ Assume P is irreducible. Then:
- ▶ Perron Frobenius $\rightarrow \rho(P) = 1$ is an eigenvalue and associated eigenvector has positive entries.
- ▶ Probabilities are obtained by scaling π by its sum.
- ▶ Example: One of the 2 models used for page rank.

Example: A college Fraternity has 50 students at various stages of college (Freshman, Sophomore, Junior, Senior). There are 6 potential stages for the following year: Freshman, Sophomore, Junior, Senior, graduated, or left-without degree. Following table gives probability of transitions from one stage to next

To From	Fr	So.	Ju.	Sr.	Grad	lwd
Fr.	.2	0	0	0	0	0
So.	.6	.1	0	0	0	0
Ju.	0	.7	.1	0	0	0
Sr.	0	0	.8	.1	0	0
Grad	0	0	0	.75	1	0
lwd	.2	.2	.1	.15	0	1

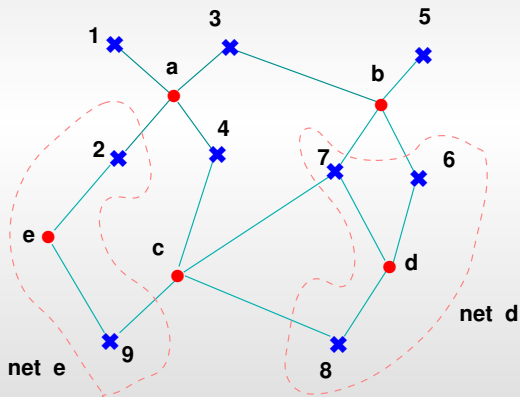
 **13** What is P ? Assume initial population is $x_0 = [10, 16, 12, 12, 0, 0]$ and do a follow the population for a few years. What is the probability that a student will graduate? What is the probability that s/he leaves without a degree?

A few words on hypergraphs

- ▶ Hypergraphs are very general.. Ideas borrowed from VLSI work
- ▶ Main motivation: to better represent communication volumes when partitioning a graph. Standard models face many limitations
- ▶ Hypergraphs can better express complex graph partitioning problems and provide better solutions.
- ▶ Example: completely nonsymmetric patterns ...
- ▶ .. Even rectangular matrices. Best illustration: Hypergraphs are ideal for **text data**

Example: $V = \{1, \dots, 9\}$ and $E = \{a, \dots, e\}$ with

$a = \{1, 2, 3, 4\}$, $b = \{3, 5, 6, 7\}$, $c = \{4, 7, 8, 9\}$,
 $d = \{6, 7, 8\}$, and $e = \{2, 9\}$

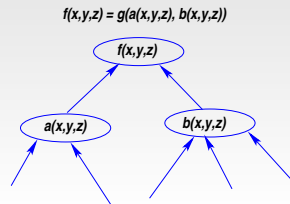


Boolean matrix:

	1	2	3	4	5	6	7	8	9	
1	1	1	1	1						a
			1		1	1	1			b
				1			1	1	1	c
						1	1	1		d
		1							1	e

A few words on computational graphs

- Computational graphs: graphs where nodes represent computations whose evaluation depend on other (incoming) nodes.



- Consider the following expression:

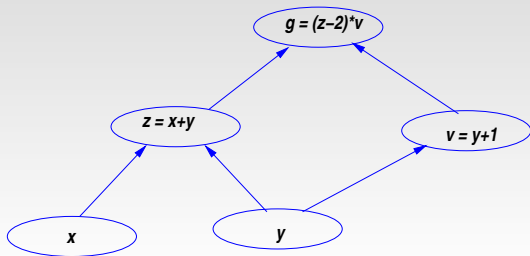
$$g(x,y) = (x + y - 2) * (y + 1)$$

- We can decompose this as
$$\begin{cases} z = x + y \\ v = y + 1 \\ g = (z - 2) * v \end{cases}$$

✦ Computational graph $\rightarrow$

✦ Given x, y we want:

- (a) Evaluate the nodes and
- (b) derivatives w.r.t x, y



(a) is trivial - just follow the graph up - starting from the leaves (that contain x and y)

(b): Use the chain rule – here shown for x only
using previous setting

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial a} \frac{\partial a}{\partial x} + \frac{\partial f}{\partial b} \frac{\partial b}{\partial x}$$

 **14** For the above example compute values and derivatives at all nodes when $x = -1, y = 2$.

Back-Propagation

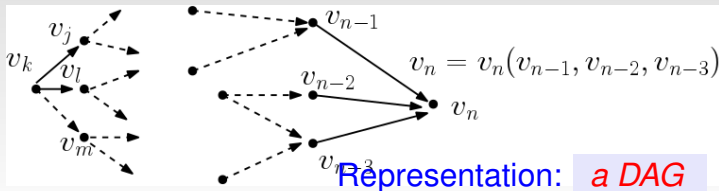
- ▶ Often we want to compute the gradient of the function at the root, once the nodes have been evaluated
- ▶ The derivatives can be calculated by going backward (or down the tree)
- ▶ Here is a very simple example from Neural Networks

$$\begin{cases} L &= \frac{1}{2}(y - t)^2 \\ y &= \sigma(z) \\ z &= wx + b \end{cases}$$



♦ Note that t (desired output) and x (input) are constant.

Back-Propagation: General computational graphs



- ▶ Last node (v_n) is the target function. Let us rename it f .
- ▶ Nodes $v_i, i = 1, \dots, e$ with indegree 0 are the variables
- ▶ Want to compute $\partial f / \partial v_1, \partial f / \partial v_2, \dots, \partial f / \partial v_e$

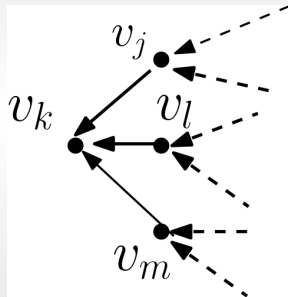
◆ Use the chain rule. $\longrightarrow$

$$\frac{\partial f}{\partial v_k} = \frac{\partial f}{\partial v_j} \frac{\partial v_j}{\partial v_k} + \frac{\partial f}{\partial v_l} \frac{\partial v_l}{\partial v_k} + \frac{\partial f}{\partial v_m} \frac{\partial v_m}{\partial v_k}$$

✦ Let $\delta_k = \frac{\partial f}{\partial v_k}$ (called ‘errors’). Then

$$\delta_k = \delta_j \frac{\partial v_j}{\partial v_k} + \delta_l \frac{\partial v_l}{\partial v_k} + \delta_m \frac{\partial v_m}{\partial v_k}$$

- ▶ To compute the δ_k ’s once the v_j ’s have been computed (in a ‘forward’ propagation) – proceed backward.
- ▶ $\delta_j, \delta_l, \delta_m$ available and $\partial v_i / \partial v_k$ computable. Node $\delta_n \equiv 1$.
- ▶ However: Must follow a **topological order** in order to obey dependencies.



Example:

